

AO LABS TECHNICAL PAPER / MAY 2026

Yum: preference-shaped media walls for taste memory

A compact system for turning cars, food, alcohol, K-pop girls, online candidates, model-assisted ranking, and cross-device preference memory into an endless visual wall that keeps becoming more personal.

yum.aolabs.io

static wall + Railway API

cars / food / alcohol / K-pop girls

fallback-first curation

Abstract

Yum is a personal media wall built around visual taste rather than generic recommendation feeds. It starts with a curated static archive of cars, food, alcohol, and K-pop girls, then extends the wall through online candidate discovery, local filters, optional AI ranking, and persistent preference memory.

The central design constraint is low-friction delight: the page should open directly into a rich wall, keep loading without a control-heavy interface, avoid obvious repeats, and preserve negative feedback across devices when the backend is available. This paper documents the system shape behind that behavior.

System Goal

Yum is a taste-memory surface: a place where cars, food, alcohol, and K-pop girls can live in one continuous visual language.

Immediate feeling

The first screen is the wall itself. Captions and sources stay quiet until interaction so the experience stays media-first.

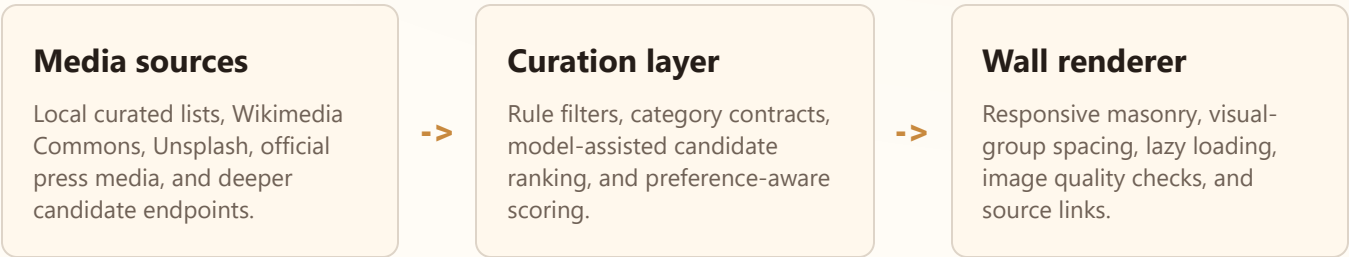
Personal curation

The feed uses fixed taste rules, category balance, source de-duplication, hidden-key memory, and sample-based scoring.

Operational resilience

The static page remains useful even when the AI endpoint or remote preference service is unavailable.

Architecture



LAYER	CURRENT IMPLEMENTATION
Public surface	GitHub Pages serves a static front end at yum.aolabs.io.
Backend	A Node service on Railway exposes curation, preference, and candidate endpoints.
AI ranking	The /api/curate endpoint can send image candidates and taste memory to the configured OpenAI model.
Persistence	Preference state is mirrored locally and stored remotely as hidden keys plus compact positive and negative samples.

Feed Contract

Yum uses an explicit feed contract instead of letting whatever source has the most available images dominate the wall. The current feed is built around cars, food, alcohol, and K-pop girls, with rules that keep the wall varied instead of letting one source or one look dominate.

1. Food and alcohol items carry the craving and memory-wall function: restaurants, dishes, drinks, screenshots, and things worth remembering.
2. K-pop girl items are intentionally softened and filtered to avoid heavy stage/editorial energy.
3. Car items are constrained toward modern compact European sedan references and away from old, show-floor, SUV, traffic, or disliked color/model patterns.
4. Source keys, visual groups, person groups, and recent placement history prevent the wall from looking like a block of repeats.

This is why the system is larger than it looks: a simple wall would just append images. Yum has to preserve taste, variety, quality, source traceability, and scroll continuity at the same time.

Curation Stack

MECHANISM	PURPOSE
Static seed pools	Guarantee that the wall starts with known-good material before network sources respond.
Blocked term filters	Reject low-fit topics such as old cars, show-floor imagery, vegetables-only food shots, low-quality portraits, and unwanted car variants.
Preference features	Convert hidden and kept examples into lightweight taste signals that adjust future ranking.
AI candidate ranking	When available, image candidates are ranked by a model using both the general taste rules and recent preference memory.
Fallback candidate generation	When a source exhausts, the system rotates through generated source templates so the wall keeps moving.

Preference Memory

Yum stores taste feedback as compact data instead of large behavioral logs. Hidden source keys remove exact items. Hidden samples act as stronger negative examples. Nearby visible items become weaker positive examples because they were present at the moment of rejection and were not removed.

The browser keeps a local mirror so the wall can respond immediately. The Railway service stores the same preference state so the memory can carry across computers and sessions when the API is reachable.

Layout and Quality Control

The wall uses responsive masonry columns with shape-aware placement. Each tile has a declared shape, source link, caption, focus point, and visual group. K-pop images are checked after load for minimum resolution before they are allowed to remain in the wall. The layout penalizes repeated people and repeated car groups so the page feels intentionally composed rather than dumped from a folder.

That composition layer is what makes the site feel like a memory surface rather than an image search result.

Limits and Next Work

1. Remote curation quality depends on the configured model and available source candidates.
2. Preference memory is compact by design, so it captures directional taste better than nuanced written explanations.
3. The wall should keep exposing the paper and system state without turning the main page into a dashboard.
4. Future improvements should add clearer source-health and preference-health readouts while preserving the first-screen media experience.